

Meta-analysis

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1 Introduction

The study by Yan et al. (2014) aims to assess the association between breastfeeding and the risk of childhood obesity through a meta-analysis of 25 observational studies. The outcome considered is obesity in childhood, defined using body mass index (BMI), while the exposure concerns different breastfeeding practices.

The association was quantified using the Adjusted Odds Ratio (AOR), that is, the OR adjusted for potential confounding variables such as maternal smoking or socioeconomic status. The main results show a significant association between breastfeeding and a reduced risk of childhood obesity. The effect estimate is 0.78, with a 95% confidence interval of [0.74, 0.81], suggesting an approximately 22% reduction in the odds of childhood obesity among breastfed children.

2 Frequentist meta-analysis

The first analyses were conducted using a frequentist approach. Section 2.1 describes the fixed-effect model, which assumes that all studies estimate a single common true effect. Section 2.2 presents the random-effects model, which also accounts for between-study variability. Finally, Section 2.3 presents a meta-regression aimed at reducing the proportion of variance attributable to heterogeneity.

2.1 Fixed-effect model

The forest plot in Figure 1 shows that most studies indicate a protective effect of breastfeeding against childhood obesity. Although the direction of the effect is consistent across nearly all studies, the individual estimates are not all centred on the same value, and some confidence intervals are particularly wide. This suggests the presence of between-study heterogeneity.

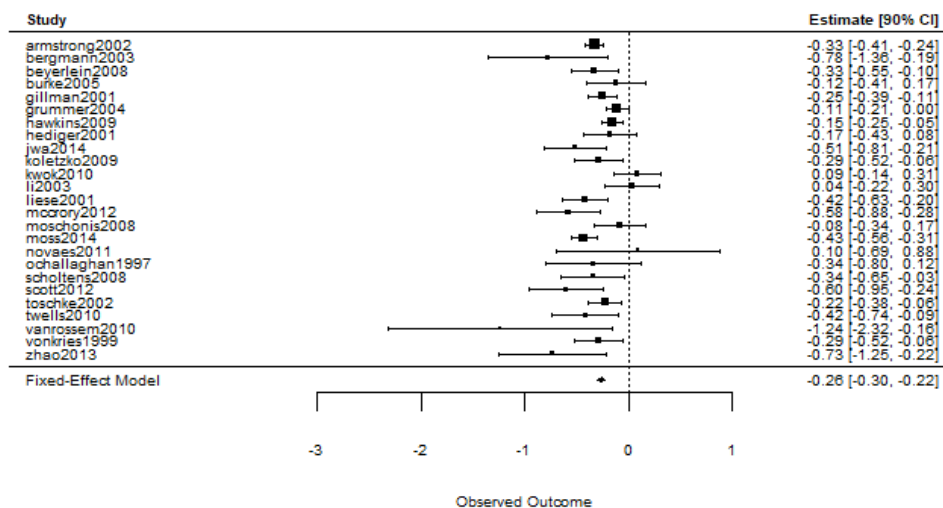


Figure 1: Forest plot for the fixed-effect model; effects are reported on the log scale.

The model results are reported in Table 1.

Statistic	Value
Number of studies (k)	25
I^2 (total heterogeneity / total variability)	46.04%
H^2 (total variability / sampling variability)	1.85
<i>Test for heterogeneity</i>	
$Q(df = 24)$	44.4743
p-value	0.0067
<i>Model results</i>	
Estimate	-0.2591
Standard error (SE)	0.0232
z-value	-11.1499
p-value	< .0001
95% CI lower bound	-0.2974
95% CI upper bound	-0.2209

Table 1: Fixed-effect meta-analysis.

The estimated effect is $e^{-0.2591} \approx 0.77$: breastfed children have approximately 23% lower odds of obesity than non-breastfed children. The confidence interval on the log scale does not include 0; therefore, the estimate is statistically significant.

Forty-six percent of the observed variability is attributable to genuine differences between studies. The ratio of total variability to sampling variability is $H^2 = Q/df = 1.85$, which measures how much the total variability observed across studies exceeds that expected from sampling error alone.

2.2 Random-effects model

Figure 2 presents the forest plot. Compared with the previous model, the weights assigned to the individual studies are more similar.

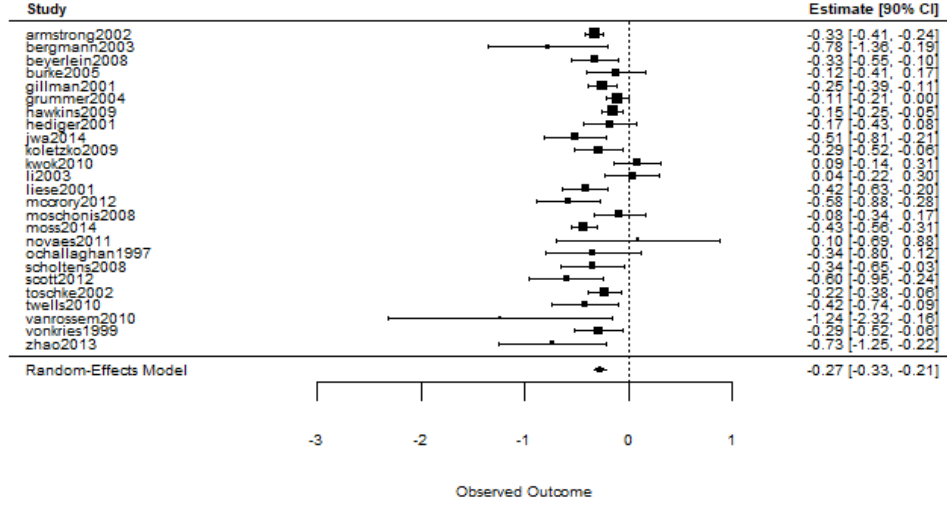


Figure 2: Frequentist random-effects analysis; effects are reported on the log scale.

Statistic	Value
Number of studies (k)	25
τ^2 (between-study variance)	0.0123
$SE(\tau^2)$	0.0087
τ	0.1107
I^2	46.04%
H^2	1.85
<i>Test for heterogeneity</i>	
$Q(df = 24)$	44.4743
p-value	0.0067
<i>Model results</i>	
Estimate	-0.2711
Standard error (SE)	0.0367
z-value	-7.3787
p-value	< .0001
95% CI lower bound	-0.3315
95% CI upper bound	-0.2106

Table 2: Random-effects meta-analysis.

The estimated mean effect is $e^{-0.2711} \approx 0.76$. Again, the estimate is significant and indicates a protective effect of breastfeeding against obesity. The between-study variance, estimated using the DerSimonian–Laird method and denoted by τ^2 , is 0.0123. Compared with the fixed-effect model, the confidence intervals are wider, as expected because an additional source of uncertainty has been included.

Figure 3 compares the normalised weights assigned to the individual studies under the fixed-effect and random-effects models. Points close to the diagonal identify studies receiving similar weights in both models. Most points lie near the line of equality; however, studies receiving a larger weight (greater precision) in the fixed-effect model tend to receive a slightly smaller weight in the random-effects model. Including τ^2 reduces extreme differences among study weights, decreasing the influence of the most precise estimates and increasing that of estimates with greater variability.

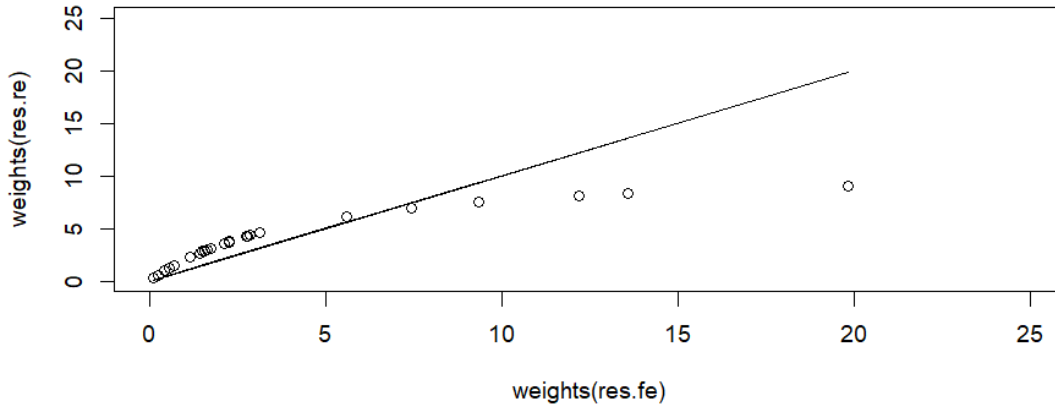


Figure 3: Comparison of weights in the fixed-effect and random-effects models.

2.3 Meta-regression

Table 3 compares the I^2 values obtained from meta-regression models including moderators for study design, publication year, outcome definition, country, and region. The I^2 values remain similar, indicating that none of the covariates explains a substantial proportion of the heterogeneity.

Among the models considered, the one including publication year is associated with the lowest residual heterogeneity, although the reduction is minimal. Figure 4 displays the meta-regression line, which shows a slight negative trend in the estimated effect as publication year increases. Some studies are widely dispersed around the line and have effect estimates substantially different from the predicted values, indicating considerable residual heterogeneity.

Model	I^2
Random-effects DL	46.04%
Meta-regression: design	50.51%
Meta-regression: year	45.99%
Meta-regression: region	47.99%
Meta-regression: outcome	49.82%
Meta-regression: country	49.81%

Table 3: Comparison of I^2 across the meta-regression models.

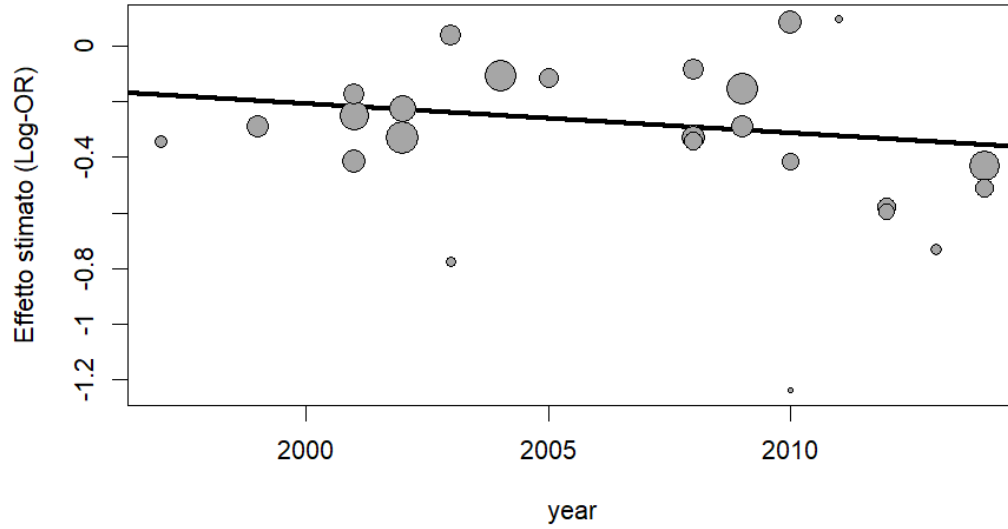


Figure 4: Meta-regression using publication year as a covariate.

Table 4 reports the estimate of the coefficient for publication year. The confidence interval includes 0, so there is no evidence of a significant association between publication year and the effect size observed in the included studies.

Predictor	Estimate	SE	z-value	p-value	95% CI lower	95% CI upper
Intercept	20.8729	15.7915	1.3218	0.1862	-5.1018	46.8477
Year	-0.0105	0.0079	-1.3390	0.1806	-0.0235	0.0024

Table 4: Estimate of the coefficient for publication year.

3 Supplementary analyses

The following analyses were performed using the random-effects model.

3.1 Trim-and-fill analysis

To assess potential publication bias, a trim-and-fill analysis was performed. Figure 5 shows that three missing studies were added to make the funnel plot symmetric.

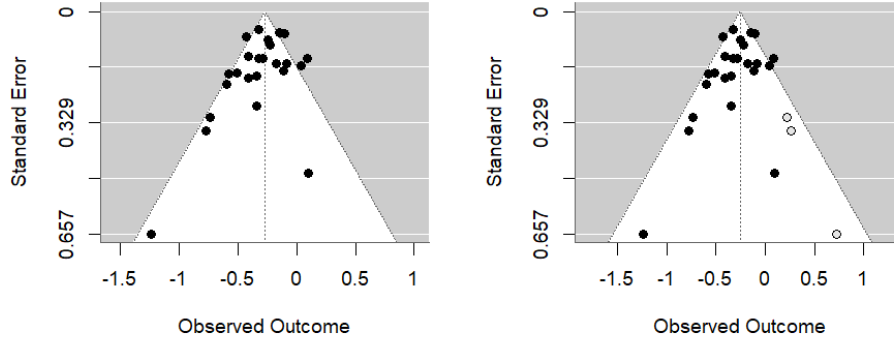


Figure 5: Trim-and-fill analysis.

The results obtained after adding the missing studies are reported in Table 5.

Statistic	Value
<i>Trim-and-fill procedure</i>	
Estimated missing studies (right side)	3
SE of missing-study estimate	3.3294
<i>Random-effects model (DL)</i>	
Number of studies (k)	28
τ^2	0.0143
$SE(\tau^2)$	0.0096
τ	0.1195
I^2	47.26%
H^2	1.90
<i>Test for heterogeneity</i>	
$Q(df = 27)$	51.1905
p-value	0.0033
<i>Adjusted model results</i>	
Estimate	-0.2577
Standard error (SE)	0.0377
z-value	-6.8362
p-value	< .0001
95% CI lower bound	-0.3315
95% CI upper bound	-0.1838
<i>Egger regression test for funnel-plot asymmetry</i>	
z-value	-1.8063
p-value	0.0709

Table 5: Trim-and-fill analysis.

Egger's test is significant at the $\alpha = 10\%$ level. After the analysis, the effect estimate is $e^{-0.2577} \approx 0.77$, similar to the previous estimate and therefore robust to publication bias.

3.2 Cumulative analysis

Figure 6 shows the forest plot for the cumulative analysis ordered by publication year. The effect estimate is progressively updated as each study is added in chronological order. The initial estimates are more variable and have wider confidence intervals. As studies are progressively added, the overall estimate stabilises and the confidence intervals narrow. The protective association between breastfeeding and obesity remains stable over time and does not appear to be affected by the inclusion of more recent individual studies.

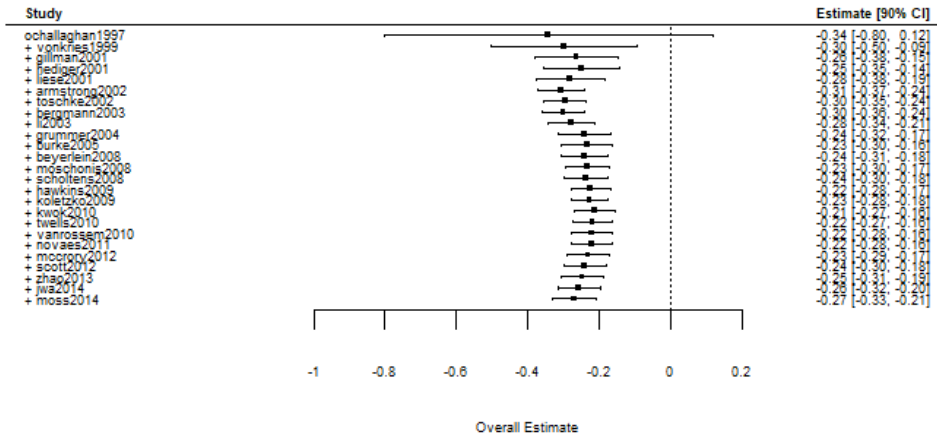


Figure 6: Cumulative analysis by publication year.

3.3 Leave-one-out analysis

Figure 7 presents the forest plot for the leave-one-out analysis, conducted by removing one study at a time and recalculating both the heterogeneity index I^2 and the effect estimate. Excluding any single study does not substantially change the overall result. The pooled estimates remain very stable, around -0.26 to -0.28 on the log scale, and the confidence intervals largely overlap, indicating that no study is particularly influential. Moreover, heterogeneity remains essentially unchanged, between 40% and 48%.

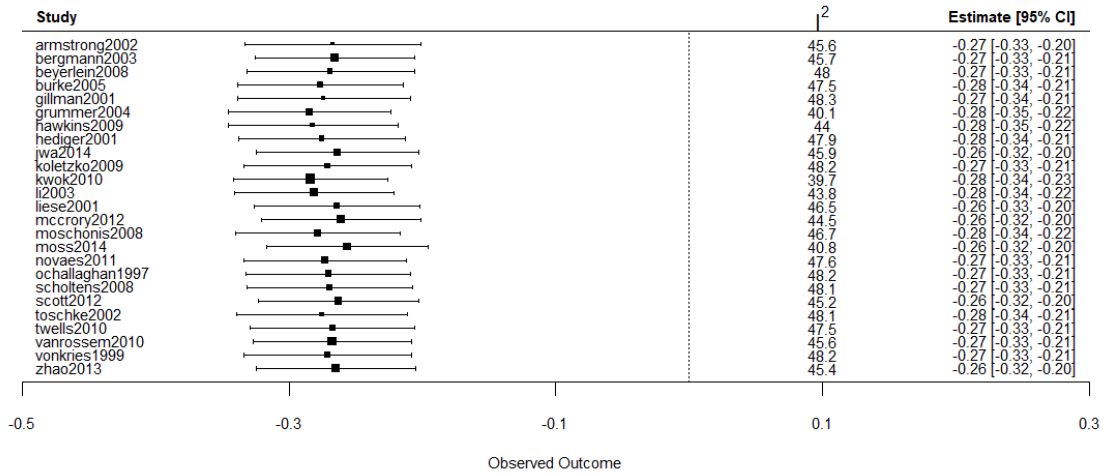


Figure 7: Leave-one-out analysis.

4 Bayesian meta-analysis

In the fixed-effect model, the overall mean effect is normally distributed as follows:

$$\beta \sim N(0, 10^6).$$

In the random-effects model, the study-specific effects β_i were modelled as

$$\beta_i \sim N(\beta, \tau^2), \quad \tau^2 \sim \text{Gamma}(0.001, 0.001).$$

In the meta-regression model, a normal prior was introduced for the regression coefficient:

$$\gamma \sim N(0, 10^4).$$

For all models, three independent MCMC chains were used and initialised with different parameter values. This makes it possible to assess convergence: if chains initialised at different points converge to the same stationary distribution, this provides evidence of stable posterior estimates. Convergence was assessed using $\hat{R}$, which was close to 1 for all parameters.

For every model, 15,000 iterations were run for each of the three MCMC chains. The first 5,000 iterations of each chain were discarded as burn-in, with a thinning interval of 1. The total number of retained posterior samples was therefore $(15,000 - 5,000) \times 3 = 30,000$.

4.1 Fixed-effect model

Figure 8 shows the posterior distribution of the overall mean parameter under the fixed-effect model.

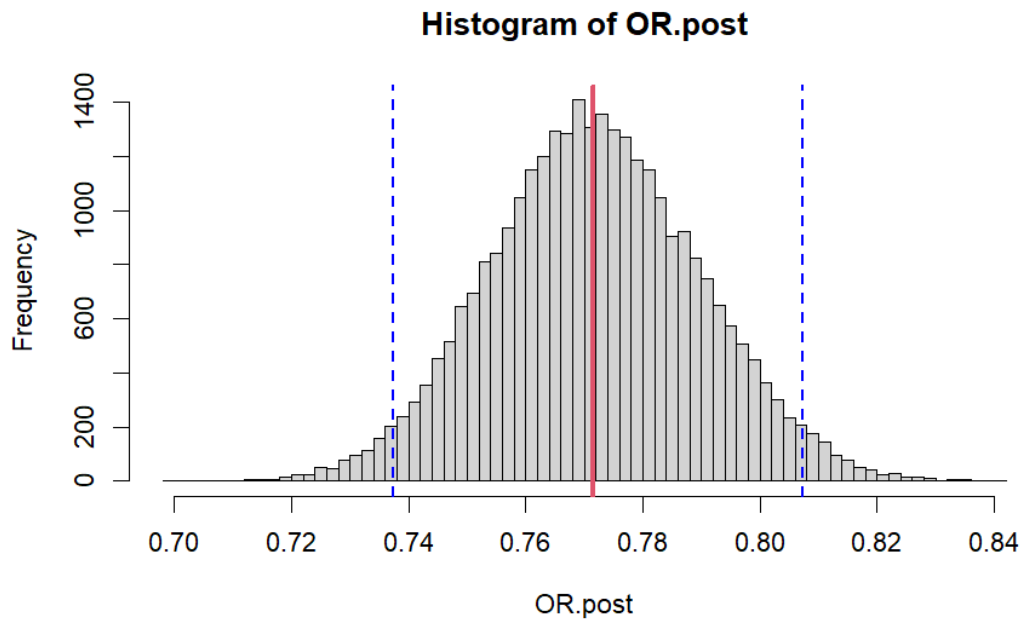


Figure 8: Bayesian estimate of the effect under the fixed-effect model. The red line indicates the posterior mean and the dashed blue lines indicate the 95% credible interval.

The effect estimate is $e^{-0.259} \approx 0.77$, with a 95% credible interval of $[0.74, 0.81]$.

4.2 Random-effects model

Figure 9 shows the posterior distribution of the overall mean parameter on the right and the posterior distribution of I^2 on the left.

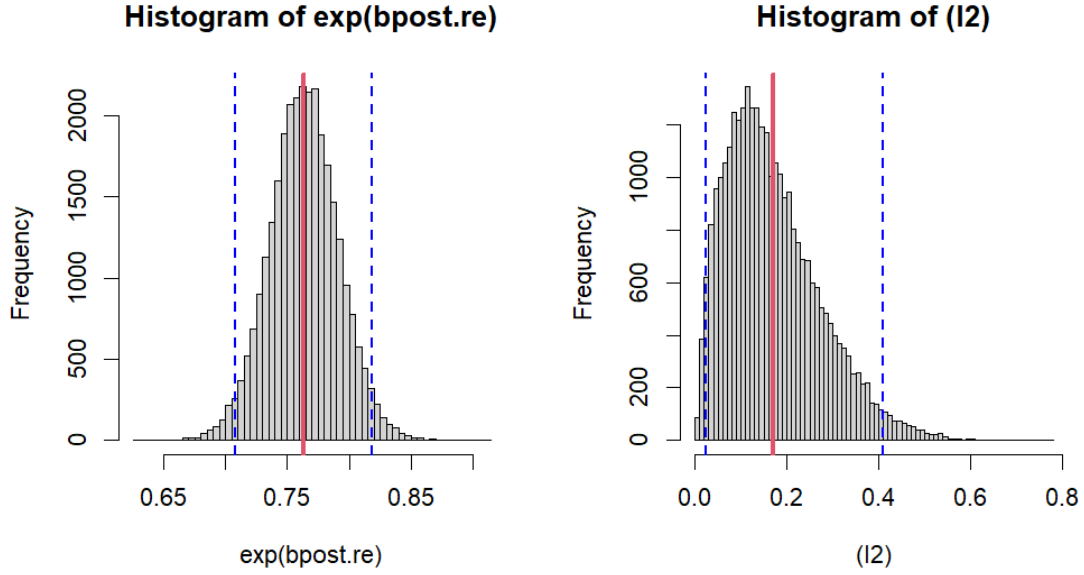


Figure 9: Estimate of the overall mean under the random-effects model. The red line indicates the posterior mean and the dashed blue lines indicate the 95% credible interval.

The estimated posterior parameters are reported in Table 6.

Parameter	Mean	2.5%	97.5%
β	0.763	0.7081	0.8185
τ^2	1.013	0.0014	0.0385
β_{pred}	0.763	0.5975	0.9663
I^2	0.170	0.025	0.410

Table 6: Posterior means and 95% credible intervals for parameters estimated by the random-effects model.

The credible interval for the posterior distribution of the overall mean is wider than that obtained under the fixed-effect model because an additional component of uncertainty, τ^2 , has been included. The 95% credible interval indicates that heterogeneity ranges from very low to moderate. Figure 10 shows selected posterior distributions of the study-specific effects.

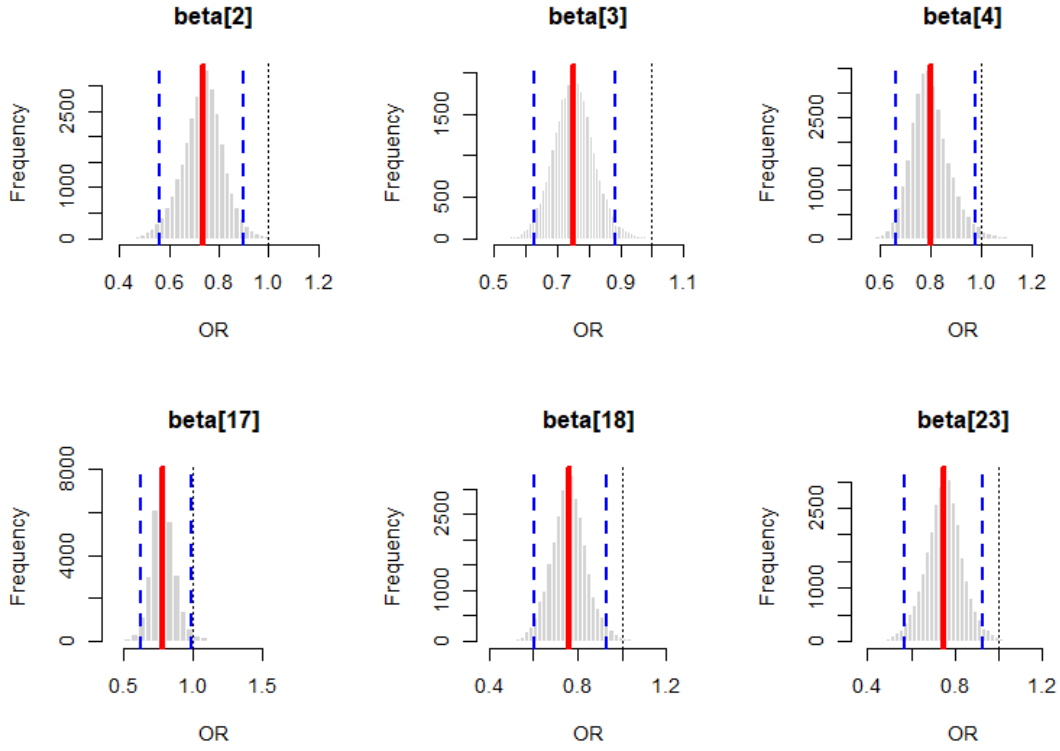


Figure 10: Histograms of the study-specific effects for studies 2, 3, 4, 17, 18, and 23. Red indicates the posterior mean and blue indicates the 95% credible intervals.

Figure 11 shows the posterior densities of the overall mean effect and the predicted effect. The black curve represents the posterior distribution of the mean effect, while the red curve represents the posterior predictive distribution. The latter is more dispersed because it accounts for uncertainty in the estimate of the mean effect as well as between-study heterogeneity. The credible interval for the predicted overall mean is wider, but it suggests that the treatment is highly likely to retain a protective effect in a future study.

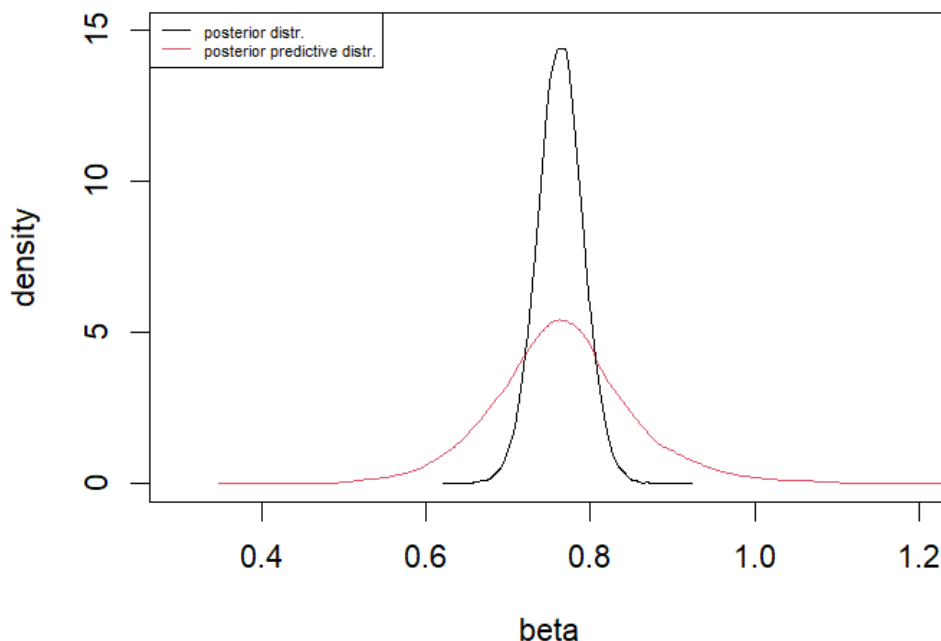


Figure 11: Comparison between the posterior distribution of the effect and the posterior predictive distribution.

The posterior probability that the overall mean treatment effect is beneficial is 1, indicating that the treatment almost certainly has a protective effect. Likewise, the posterior predictive probability that a new study will show a beneficial effect is 0.984, suggesting a high probability that the treatment effect in a future study will remain below 1, even after accounting for between-study heterogeneity.

4.3 Meta-regression

A meta-regression was estimated using publication year as a covariate. The estimated parameters are reported in Table 7.

Parameter	Mean	2.5%	97.5%
β	0.846	0.721	0.989
γ	-0.014	-0.034	0.005
τ^2	0.013	0.002	0.039
I^2	0.176	0.028	0.416

Table 7: Posterior summary of the Bayesian meta-regression model.

The credible interval for the regression coefficient γ includes zero; therefore, there is no significant evidence of an association between the covariate and the observed effect.

The estimated effect is larger than under the random-effects or fixed-effect model, suggesting a weaker protective effect after adjustment for the covariate included in the model. The posterior estimate of residual heterogeneity does not change substantially compared with the random-effects model, suggesting that the covariate does not explain the heterogeneity across studies.

Figure 12 shows the posterior distributions of the overall effect adjusted for the covariate and of the regression coefficient.

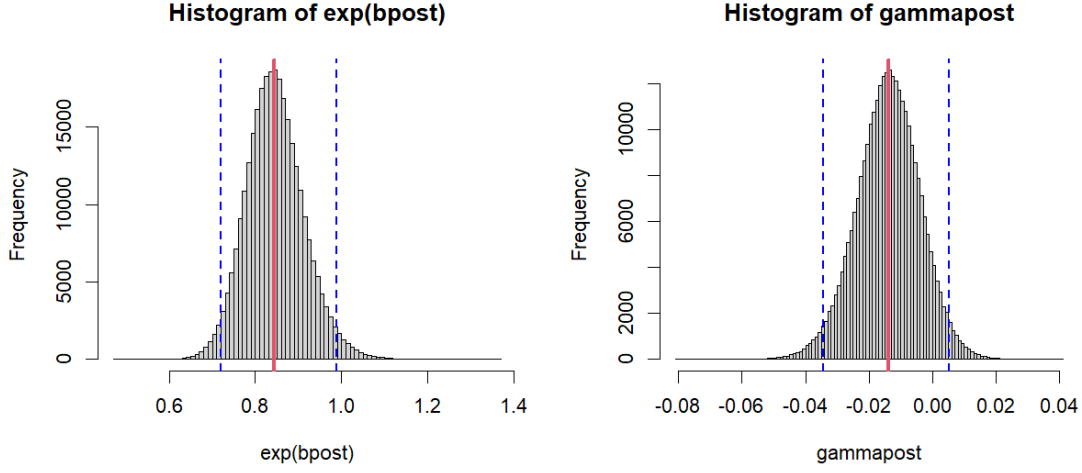


Figure 12: Posterior distributions of the overall effect and the regression coefficient.

Figure 13 compares the posterior distribution of I^2 obtained from the random-effects model with that obtained after introducing the covariate in the meta-regression. The two distributions almost completely overlap, indicating that the covariate does not explain a meaningful proportion of the heterogeneity across studies.

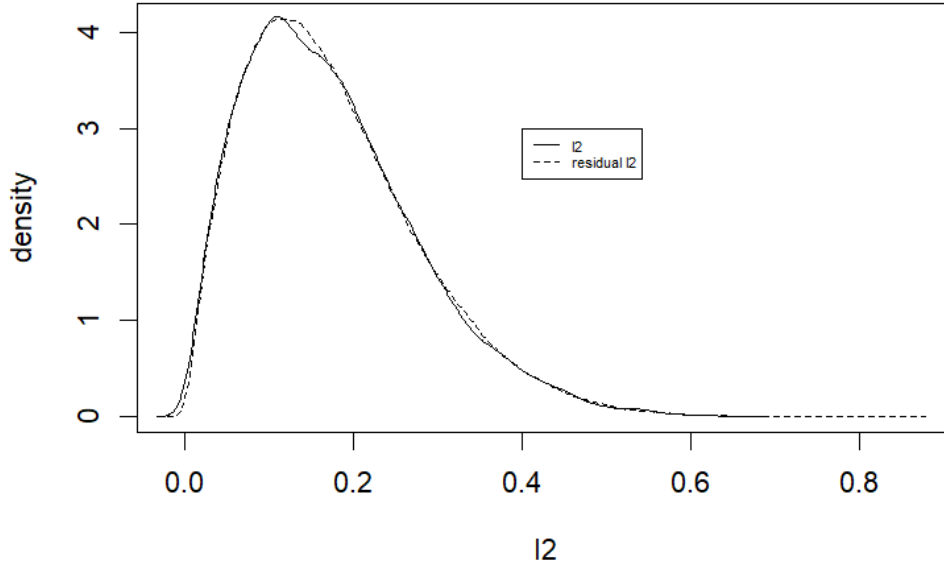


Figure 13: Comparison of the I^2 posterior distributions under the random-effects and meta-regression models.

References

Jing Yan, Lin Liu, Yun Zhu, Guowei Huang, and Peizhong Peter Wang. The association between breastfeeding and childhood obesity: a meta-analysis. *BMC Public Health*, 14(1267), 2014. doi: [10.1186/1471-2458-14-1267](https://doi.org/10.1186/1471-2458-14-1267).